Typesetting N’ko Mathematical Equations in LibreOffice

Overview

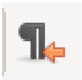
N’ko is the writing system used primarily for the Manding languages. The script runs completely right-to-left, unlike Arabic where numbers are composed left-to-right. Proper typesetting of N’ko mathematical equations requires a font that supports mirroring of mathematical symbols like XITS Math Extended as well as a type setting program that supports right-to-left formula composition like LibreOffice.

Mathematical Conventions

	Example	**Latin**	**(N’ko) ߒߞߏ**
Digits		1 2 3 4 5 6 7 8 9 0	߀ ߁ ߂ ߃ ߄ ߅ ߆ ߇ ߈ ߉
Letters		a-z/A-Z	ߊ-ߦ
Parameter/ Coefficients	ax + by	a-d	ߊ-ߋ
Cartesian Coordinates		x, y, z	ߛ߸ ߜ߸ ߝ
Length/Width		l/w	ߟ\ߥ
Radius/Diameter	d = 2r	r/d	ߘ\ߙ
Angles	cos(θ)	$\alpha, \beta, \gamma, \theta, \varphi$	ߊ߸ ߎ߸ ߌ߸ ߍ߸ ߐ
Functions	f(x)	f, g, h	ߓ߸ ߔ߸ ߕ
Decimal Separator	1.25	.	.
Thousands Separator	1,250	,	߸
Mean		μ	ߡ
Integer Variables/Indices	$\sum_{i=1}^{n} i$	i-k	ߐ-ߞ
Fixed Integer/Count/ Probability	$\lim_{n \to \infty} \left(1+\frac{1}{n}\right)^n$	n	ߣ

Typesetting

LibreOffice supports the mirroring of all mathematical symbols. This handles the majority of cases as long as the formula editor is set to Right-to-Left mode. This is done by looking for the following button.



For example the simply by placing the editor in Right-To-Left mode the following formulas automatically mirror.

LTR	RTL
$\square < \square$	$\square > \square$
$\square \leqslant \square$	$\square \geqslant \square$
$\sqrt[\square]{\square}$	$\reflectbox{$\sqrt[\square]{\square}$}$
$\square \in \square$	$\square \ni \square$
$\int_{\square} \square$	$\square \reflectbox{$\int$}_{\square}$
$\oiint_{\square} \square$	$\square \reflectbox{$\oiint$}_{\square}$

Functions with specific names have been added to the user interface. Below is a complete list of those that have been included.

Logorithms/Exponentials

Latin	N'ko
$e^{\square}$	ߋ$^{\square}$
$\ln(\square)$	ߡߟߐ$(\square)$
$\exp(\square)$	ߖߛߋ$(\square)$
$\log(\square)$	ߡߟ$(\square)$

Limits

Latin	**N'ko**
lim ❑	❑ [illegible]
lim inf ❑	❑ [illegible]
$\overset{❑}{\text{lim}}$ ❑	❑ [illegible]

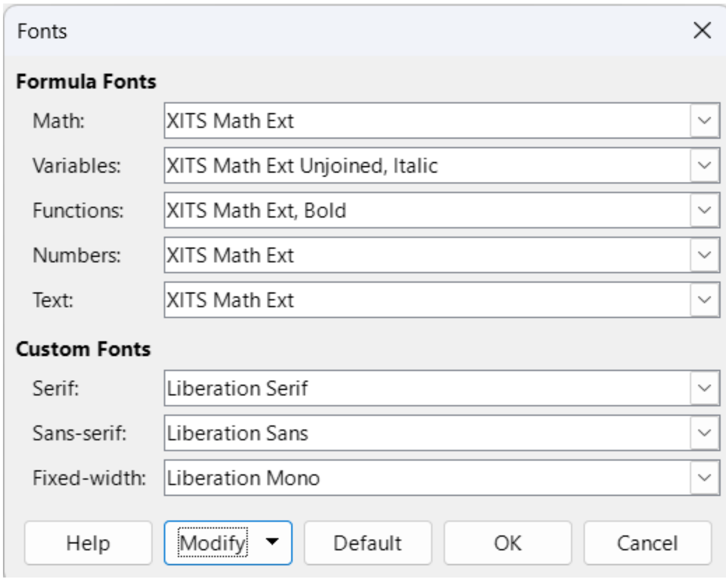
Trigonometry

Latin	**N'ko**
sin (❑)	(❑) [illegible]
cos (❑)	(❑) [illegible]
tan (❑)	(❑) [illegible]
sec (❑)	(❑) [illegible]
csc (❑)	(❑) [illegible]
cot (❑)	(❑) [illegible]
sinh (❑)	(❑) [illegible]
cosh (❑)	(❑) [illegible]
tanh (❑)	(❑) [illegible]
coth (❑)	(❑) [illegible]
sech (❑)	(❑) [illegible]
csch (❑)	(❑) [illegible]
arcsin (❑)	(❑) [illegible]
arccos (❑)	(❑) [illegible]
arctan (❑)	(❑) [illegible]

Font Choices

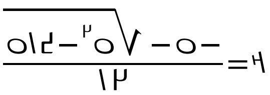
XITS Math Extended includes 5 fonts: XITS Math Ext Regular, XITS Math Ext Bold, XITS Math Ext Italic, XITS Math Ext Unjoined Regular, and XITS Math Ext Unjoined Italic. For typesetting N'ko it is important to distinguish between variables and functions. Therefore variables should use the Unjoined variations. This ensures that two variables next to one another can be read as distinct. Though it is not necessary setting functions a Bold font can help them be more recognizable.

LibreOffice provides the ability to select the font for different elements of a formula. Below is an optimal setup. The bold font not necessary for legible equations but can be used if needed.

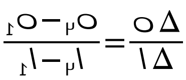


Sample Formulas

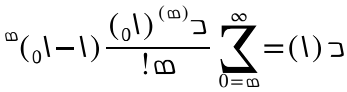
Quadratic Equation:



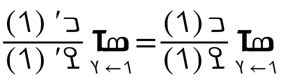
Slope Formula:



Taylor Series:



L’Hôpital’s Rule:



L’Hôpital’s Rule with Trigonometry: